Learned Today

Date: 7 March 2026		
Room	Terms	wdym man
OSI Model	OSI Model	Open systems interconnection model, how data is transported between devices, it exist in 7 layers 7. Application 6. Presentation 5. Session 4. Transport 3. Network 2. Data link 1. Physical *Why hackers care?* *Different attacks target different layers.* *Example: spoofing IP addresses targets Layer 3 (Network).*

Layers	#	*huh?*
Application	7	where user applications interact with the network. (GUI) ex. *When you type a website into a browser, this is the layer the browser interacts first (browser = application)*
Presentation	6	how data is formatted or encoded, this layer acts as a translator *Formatting / encryption / compression of data*
Session	5	manages connection between devices *ex. keeping a login session active*
Transport	4	breaks data into segments and ensures reliable delivery, *via: TCP, UDP (key idea = moving data reliably.)*
Network	3	IP address *ex. routers use IP addresses to route packets. (OSPF, RIP)*
Data Link	2	Uses MAC addresses to send frames inside a local network (LAN)
Physical	1	hardware basically electrical signals, fiber/light signals, cables, raw bit transmission

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	NIC	*Network Interface Card*, handles MAC address and sending signals through hardware
	Encapsulation	pieces of info get added to the data
	OSPF & RIP	OSPF: Open Shortest Path First RIP: Routing Information Protocol
	TCP & UDP	TCP: Transmission Control Protocol *ex. emails, sharing files, internet browsing* UDP: user datagram protocol *ex. video streaming, gaming*

Notes :

- *NIC works mainly on Layer 1 and Layer 2*
- *Router: works at Layer 3 (Network) using IP addresses*
- *Switch: works at Layer 2 (Data Link) using MAC addresses*